Few-Shot Open-Set Collaborative Classification of Multispectral and Hyperspectral Images With Adaptive Joint Similarity Metric

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***Abstract*— Hyperspectral images (HSIs) have higher spectral resolution than multispectral (MS) images, but they have a narrower swath than MS images. The limited spectral resolution of MS images constrains their classification capabilities, and annotating remote sensing data is time-consuming and laborious. In addition, large-scale MS images may contain unknown classes not present in the training data. This article attempts to use partially overlapping HS images with limited labels to assist in the classification of large-scene MS images. It can correctly distinguish known classes and simultaneously identify unknown classes, thereby achieving better classification results for MS images. To address this challenge, a few-shot open-set HS–MS image collaborative classification method is proposed. Specifically, a spectral–spatial feature interactive enhancement (SSFIE) module is designed for richer feature extraction and enhanced classification capabilities in the feature extraction stage. In the few-shot learning (FSL) stage, an adaptive joint similarity metric criterion is proposed to improve feature mapping between the source and target domains. Discriminative joint probability adaptation (DJPA) is used for domain adaptation and to enhance feature discriminability, while batch nuclear-norm maximization (BNM) is employed to increase the feature diversity. In the testing phase, the open-set classification module is designed to correctly classify samples of known classes while simultaneously distinguishing unknown classes. The experimental results on four cross-domain HS–MS data pairs demonstrate that our proposed method outperforms state-of-the-art methods.**

***Index Terms*— Domain adaptation, few-shot learning (FSL), hyperspectral image (HSI), multispectral (MS) image, open-set classification.**

I. Introduction

THE hyperspectral images (HSIs) combine the spectral information of ground objects with the spatial geometric information of ground objects and can detect substances that cannot be detected in conventional bands. They have been widely used in image classification [1], [2], [3], target detection [4], [5], agricultural monitoring [6], [7] and other applications [8]. Classification technology of HS images uses continuous narrowband spectral information to classify ground objects into different classes, which has important application value. However, due to the limitations of imaging equipment, the swath of HS images is relatively narrow, making it difficult to conduct large-scale observations. Although multispectral (MS) images can make it easier to obtain images of large scenes, the limited spectral resolution of MS images limits their classification capabilities, and MS images of large scenes may contain classes outside the common coverage area. Moreover, the collection and labeling of remote sensing data are expensive and time-consuming. Therefore, this article mainly studies the cross-domain open-set collaborative classification method of HS–MS images in the case of few-shot samples. The proposed method aims to improve the classification ability of large-scene MS images by utilizing partially overlapping HS images with limited labeled samples, effectively identifying unknown classes in MS images while correctly classifying known ones.

Collaborative classification of HS–MS images can be seen as a problem of collaborative utilization of multimodal data [9]. Since the data distribution in the source domain and target domain is different, it can be solved through heterogeneous transfer learning and few-shot learning (FSL) [10]. Heterogeneous transfer learning can adapt two domains with different feature dimensions, overcome the differences between the source domain and the target domain, and achieve effective knowledge transfer. Chen et al. [11] utilized cross-domain Laplacian graphs with label information to explore the similarity between cross-domain samples of the same class. Li et al. [12] initially generated pseudo-labels for unlabeled target domain data and then employed feature distribution alignment and local information retention to learn the optimal common subspace. Tsai et al. [13] mapped the source domain features to the target domain subspace and combined standard sample selection and distribution alignment to learn the optimal feature mapping in this subspace. Wu and Ng [14] employed embedding matrices to embed each domain into multiple graphs, thereby preserving the local structural information of each domain. Zhang and Wu [15] proposed the discriminative joint probability maximum mean discrepancy method to reduce feature differences by adapting the joint probability distribution between the two domains.

Received 19 May 2024; revised 14 September 2024 and 20 October 2024; accepted 30 October 2024. Date of publication 19 November 2024; date of current version 10 December 2024. This work was supported in part by the Distinguished Young Scholars of Natural Science Foundation of China under Grant 62025107, in part by the Young Elite Scientists Sponsorship Program-break by China Association for Science and Technology (CAST) under Grant 2022QNRC001, and in part by the Key Research and Development Program of Heilongjiang Province under Grant JD22A019. *(Corresponding author: Tianzhu Liu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3502236

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The migration neural tree method [16] mapped heterogeneous data to common feature space and designed a migration neural decision forest for domain adaptation. In the field of remote sensing, Hong et al. [17] improved the classification of MS images by learning the common subspace of HS–MS images while aligning the manifold structures of the two domains. Yao et al. [18] proposed the unsupervised common subspace learning method, which can learn cross-modal projections from multimodal data, thereby effectively capturing discriminative representations. Guo et al. [19] proposed the structure-preserved discriminative distribution adaptation method for HS–MS image joint classification. Specifically, this method first maps HS images and MS images to a common subspace by preserving local structures and applying discriminative least-squares constraints. This process improves the discriminability of features from different classes within the subspace, thereby improving the classification capabilities of MS images. However, it is not suitable for classification with few samples and unknown classes. Hong et al. [20] jointly learned multimodal feature representations from HS and MS superpixels, leveraging the diverse and complementary information from these modalities to achieve more accurate land cover classification results. Although these methods facilitate knowledge transfer between the two domains, they require a sufficient number of labels and identical label types in both domains to establish reliable mapping relationships.

In recent years, FSL has received more and more attention in the field of machine learning. It aims to address the challenge of learning and generalizing from a limited number of labeled samples [21]. ProtoNet [22] is a metric-based meta-learning method that mainly learns a metric space to perform classification by calculating the distance to the prototype representation of each class. MatchNet [23] extracts the feature vector and introduces an attention mechanism to analyze the similarity between the query sample and the support set. Ye et al. [24] introduced the few-shot embedding adaptation with the transformer (FEAT) method. This approach adapts instance embeddings to the target classification task through set-to-set functions, resulting in task-specific and discriminative embeddings for more effective classification. Liu et al. [25] proposed a prototype network based on cosine similarity and a bias reduction module for prototype correction to achieve FSL. To address the challenge of few-shot labeled samples in remote sensing data, scholars have utilized FSL methods for classification. Li et al. [26] presented the deep cross-domain FSL (DCFSL) method, which combines FSL and domain adaptation to classify cross-domain HS images. Based on DCFSL, Guo et al. [10] designed feature enhancement modules and subdomain alignment to achieve collaborative classification of HS–MS images. This method first maps data from both domains to the same dimension using a mapping module. It then enhances the diversity of the feature distribution through the feature enhancement module. Finally, it combines the losses from FSL and subdomain alignment for training. During testing, the MS image is input into the trained feature extractor, and classification is performed using the nearest neighbor algorithm. However, this method is only applicable to few-shot HS–MS classification in closed-set cases and is not suitable when the MS image contains unknown classes. Xi et al. [27] designed a lightweight cross-scale convolutional network and then used metric learning based on Mahalanobis distance to achieve cross-domain HS image classification. Bai et al. [28] proposed an FSL classification model based on adaptive subspace and feature transformation for classification. Zhang et al. [29] designed feature- and distribution-level cross-domain graph alignment based on graph neural networks to alleviate the impact of domain differences on FSL. Despite these advancements, most FSL methods rely on a single similarity measure, which can lead to similarity bias [30], thereby reducing the generalization ability of the model. Furthermore, many FSL classification methods fail to account for the presence of unknown classes in the test samples.

The situation where the test sample contains unknown classes is defined as open-set classification, which is more in line with the actual situation. A reliable classification system should not only correctly identify known feature classes but also reject unknown classes. In order to solve the problem of open-set classification, the open-set nearest neighbor classifier (OSNN) method [31] extends the nearest neighbor classifier, which has the ability to identify unknown class samples, making it suitable for open-set recognition. Zhang et al. [32] proposed an open-set classification method for remote sensing image scene classification, aimed at ensuring model transferability and enhancing feature discriminability. This method projects the data into a common subspace using manifold embedding and updates the pseudo-labels of the test data iteratively to identify both known and unknown classes. Although this method can distinguish unknown classes, it requires that the feature dimensions of the source and target domain data are consistent. Li et al. [33] introduced a new open-set HSI classification framework based on supervised contrastive learning. This framework applies contrastive learning to spectral and spatial feature learning to bring samples of the same class closer and distinguish unknown classes from known ones. Pal et al. [34] proposed a residual 3-D convolution block attention module to extract the discriminative prototype of each known class. Subsequently, they designed a meta-learning-based outlier calibration network to distinguish between known and unknown samples. Liu et al. [35] employed a multitask deep learning method to simultaneously perform classification and reconstruction for open-set classification of HS images. Xie et al. [36] proposed an HS open-set classification method based on feature consistency. The problem addressed in this article involves domain adaptation and the classification of unknown classes. In cross-domain open-set classification problems, the majority of current research is centered on computer vision methods. Saito et al. [37] proposed open-set domain adaptation, utilizing adversarial learning to distinguish between known and unknown classes in the target domain. Wang et al. [38] proposed the open-set locality preserving projection algorithm to reject outliers from unknown classes through threshold rejection. Fang et al. [39] proposed the distribution alignment with an open difference method, which can achieve unsupervised open-set domain adaptation. The SnaTCHer [40] method

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measures the difference between transformed prototypes and changes the unknown class distribution estimation problem into a relative feature conversion problem. However, due to certain differences between remote sensing images and natural images, existing machine vision methods cannot produce good results when directly applied to remote sensing images.

In summary, it is evident that most cross-domain few-shot classification methods overlook the situation where the test data contain unknown classes. When faced with any test sample, the algorithm is forced to classify it into one of the known classes, thereby weakening the classification performance of the model. In addition, most metric-based few-shot classification methods adopt a single similarity measure, which may introduce some similarity bias and diminish the generalization ability of the model, especially in cases with limited samples. Moreover, the majority of existing open-set classification methods are designed for natural images and ignore the characteristics of HS–MS images. Therefore, this article proposes a few-shot open-set HS–MS collaborative classification model, which mainly consists of a feature extraction module, the FSL module, the discriminative joint probability adaptation (DJPA) module, the batch nuclear-norm maximization (BNM) module, and an open-set classification module. First, the feature extraction module consists of the 3-D residual convolution block and a specially designed spectral–spatial feature interactive enhancement (SSFIE) module to generate rich spatial–spectral mapping features. Second, an FSL module based on adaptive joint similarity metric is proposed to learn the metric space of the source domain and target domain. At the same time, the DJPA module is used for domain alignment to reduce the difference between the source domain and the target domain. The BNM module is introduced to enhance feature discriminability. In the testing phase, the open-set classification module is designed to correctly classify known classes and identify unknown classes in the test data. The main contributions of this article are summarized as follows.

1) To the best of our knowledge, this work is among the first to utilize partial HS images to assist large-scale few-shot open-set MS image classification.
2) In the feature extraction module, the SSFIE module is designed to interactively enhance both the spectral information and spatial information of the image, providing rich feature representation.
3) The adaptive joint similarity metric criterion is proposed, allowing the model to learn more discriminative features from limited data, thereby improving the generalization ability of the model.
4) In order to improve the discriminability of features, the DJPA and BNM modules are introduced to further improve the classification accuracy.

This article is organized as follows. Section II presents the proposed method in detail. Section III introduces experimental data and results. Finally, Section IV concludes this article.

II. Proposed Algorithm

The method proposed in this article addresses the issue of unknown samples in the collaborative classification of MS-HS images, especially in scenarios with a limited number of labeled samples. It primarily leverages the spectral resolution capabilities of HS images within the shared coverage area to improve the classification of MS images captured over larger scenes. This approach accurately classifies MS images of extensive scenes into known classes while also identifying unknown classes not covered in the common areas. The specific method will be introduced as follows.

A. Method Overview

As shown in Fig. 1, the framework of this method consists of four main components: the feature extraction module, the FSL module, the DJPA, and the BNM module. In the feature extraction module, the process begins with two mapping layers that align sample features from the source domain HS and target domain MS data to the same dimension. Following this, a residual 3-D convolutional neural network is utilized to extract spatial–spectral features from the samples. In addition, the SSFIE module is designed to facilitate the interactive enhancement of spatial–spectral features. The source and target domains simultaneously perform FSL with an adaptive joint similarity measurement criterion. At the same time, DJPA is employed to adjust probabilistic adaptation between the source domain and target domain features to reduce the difference between the two. The BNM module is used to prevent the model from overly focusing on certain classes during the training process, thereby improving the classification accuracy. During the testing phase, an open-set classification module is designed to identify unknown classes. As shown in Fig. 2, similar to the training stage, the target domain data first extract discriminative features through the feature extraction module and then passes through the open-set classification module to correctly classify known classes and identify unknown classes. Section II-B–II-F provide a detailed explanation of the modules in the proposed framework.

B. Feature Extraction Module

Due to the different sensors used for HS and MS images, there is a difference in their spectral resolution. To solve this problem, two feature mapping layers are first employed to map the HS image and MS image to the same spectral dimension. Subsequently, a 3-D deep residual network is used to extract spatial spectral features of HS–MS data. The proposed SSFIE module is then applied to improve the extracted spatial–spectral features. The detailed network of the feature mapping layer, the feature extraction layer, and the SSFIE module is introduced as follows.

1) Feature Mapping Layer: As shown in Fig. 3, the feature mapping layer primarily consists of a 2-D convolutional layer. Its main objective is to unify the spectral dimensions of HS and MS images, which can be expressed as follows:

$$X' = X \times M \tag{1}$$

where $X \in R^{9\times 9\times \mathrm{ch}}$ represents the input MS or HS image cube, 9×9 is the spatial size of the input cube, and ch is the dimension of source domain HS data or target domain

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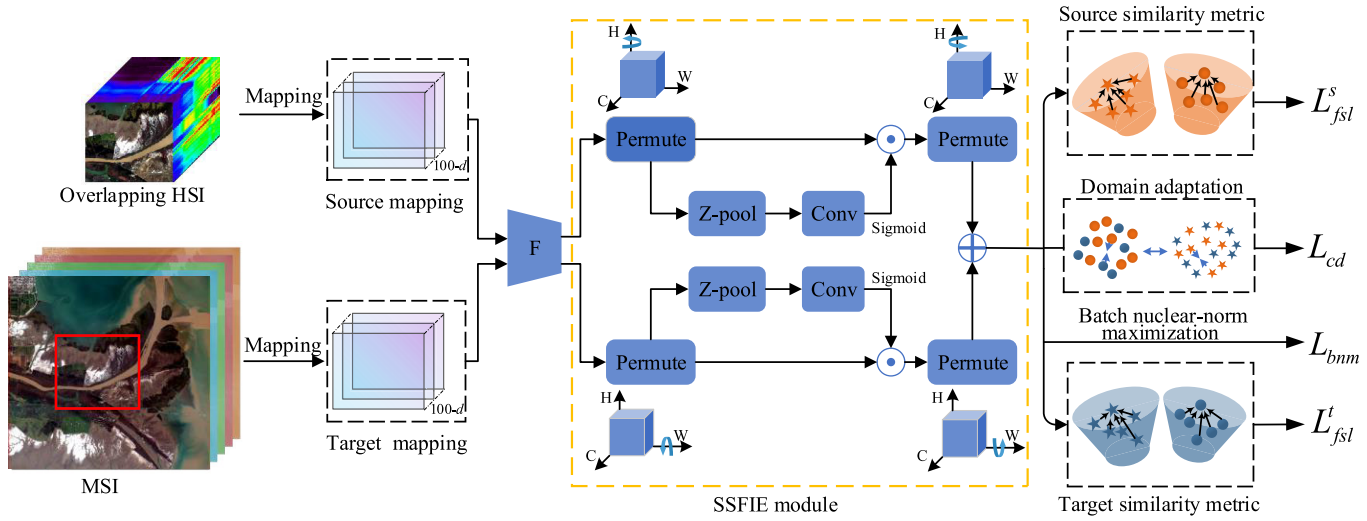


Fig. 1. Proposed model for the classification task. It contains the feature extraction module, the FSL module, the discriminative probability adaptation module, and the BNM module.

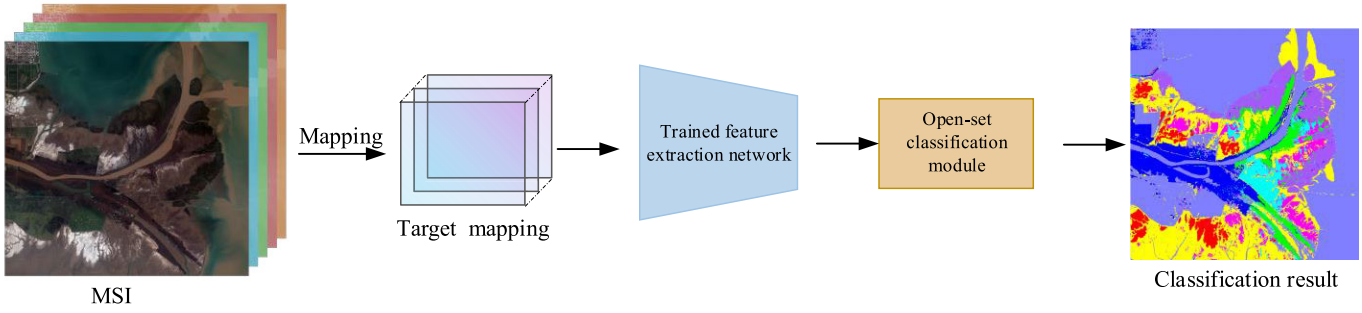


Fig. 2. Flowchart in the testing phase.

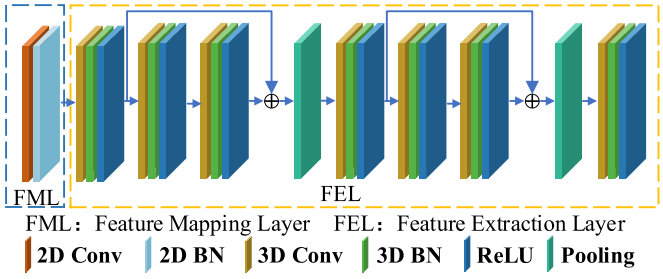


Fig. 3. Structure of the feature mapping layer and feature extraction layer.

MS data. $X' \in R^{9\times9\times100}$ is the mapped data. The mapping matrix $M \in R^{\text{ch}\times d}$ is obtained through learning, allowing MS or HS images to be mapped to the same dimension d. In the experiment, the mapped dimension d is set to 100.

2) Feature Extraction Layer: The 3-D deep residual network F is employed to extract the spatial–spectral features of the HS–MS image. The input to the feature extraction layer is the output from the feature mapping layer. This section first reviews the principles of 3DCNN and residual structure and then introduces the structure of the feature extraction layer.

3DCNN is more suitable for feature extraction of HS data than 2DCNN, as it performs convolution in the x, y, and z dimensions. In the jth feature map in the ith convolution layer, the value v_{ij}^{xyz} at (x, y, z) can be expressed as

$$v_{ij}^{xyz} = f\left(b_{ij} + \sum_{k=1}^{m}\sum_{p=0}^{P_i-1}\sum_{q=0}^{Q_i-1}\sum_{r=0}^{R_i-1} w_{ijk}^{pqr} v_{(i-1)k}^{(x+p)(y+q)(z+r)}\right) \quad (2)$$

where P_i, Q_i, and R_i represent the size of the convolution kernel in the x-, y-, and z-directions, respectively; w_{ijk}^{pqr} denotes the value of the position (p, q, r) associated with the kth feature map; b_{ij} is the bias; and $f(\cdot)$ is the activation function.

Deep neural networks have more complex architectures and are better at learning deep features of images. However, increasing the depth of the network can lead to issues such as network degradation and problems with gradient vanishing or exploding. In order to solve this problem, the connection between the next layer or the next two layers of neurons can be skipped in some feature layers of the network to weaken the strong connection between each layer. This kind of neural network is called residual neural network. Assuming that the original input of the network is x, the learned feature is $F(x)$, and the residual can be defined as

$$W(x) = F(x) - x \quad (3)$$

where $W(x)$ is the output after two convolutional layers and an activation function. The original input x is connected to

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$W(x)$ through a skip link, and the input of the next layer of the network is $W(x) + x$.

The structure of the feature extraction layer is shown in Fig. 3. In order to better extract features, the structure of the feature extraction layer is the same as in [26] and [29]. Specifically, the 3-D deep residual network F consists of two residual blocks, two maximum pooling layers, and a convolution layer. The convolution kernel size is $3 \times 3 \times 3$, and each residual block contains three 3-D convolutional layers and a shortcut connection.

3) SSFIE Module: This article designs an SSFIE module, which enhances HS–MS spectral–spatial features extracted from the feature extraction layer through two branch structures. In [41], the input HS data feature map is processed sequentially through a spectral attention module and a spatial attention module, which are arranged in series. The spectral attention module calculates the weight for each spectral channel, which is then used to scale each channel of the feature map. However, during this weight calculation, the input vector must be compressed into an intermediate vector using global average pooling or maximum pooling, which results in a loss of spatial information. Moreover, the spectral attention module and the spatial attention module operate independently, which overlooks the relationship between spectral and spatial dimensions and results in significant computational overhead. To address these limitations, this article proposes an SSFIE module to enhance the spectral–spatial feature. It utilizes two branch structures to capture the cross-dimensional interaction of HS or MS images and calculate the feature weight of the spectral–spatial interaction. The input characteristics can be represented as $z \in R^{C\times H\times W}$. This module captures the dependence between the (C, H) and (C, W) spectral–spatial mapping features of the input tensor through rotation operations. Its complete structure is shown in Fig. 1. The two branches calculate the cross-dimensional interaction between the spectral channel C and the spatial dimensions H and W, which can better integrate spectral and spatial information. In the first branch, an interaction is established between the spatial dimension H and the spectral dimension C. The input feature map $z \in R^{C\times H\times W}$ is rotated 90° counterclockwise along the H-axis to obtain $z_1 \in R^{W\times H\times C}$. After applying the Z-pool layer, the tensor is represented as $\hat{z}_1 \in R^{2\times H\times C}$. The Z-pool layer can be expressed as follows:

$$Z\text{-pool}(z) = \left[\text{MaxPool}_{0d}(z), \text{AvgPool}_{0d}(z)\right]. \tag{4}$$

The Z-pool layer is responsible for reducing the C-dimensional tensor to two dimensions and connecting the average pooled features and the maximum pooled features in this dimension. Then, the output $\hat{z}_1^* \in R^{1\times H\times C}$ is generated through convolutional layers and batch normalization layers. The spectral–spatial interaction weight w_1 is generated by the sigmoid function. Finally, the output tensor $z_1 \cdot w_1$ is rotated 90° clockwise along the H-axis to ensure that it maintains the same shape as the input tensor.

In the second branch, an interaction is established between the W dimension and the C dimension. The whole process is similar to that of the first branch, except that the input tensor $z \in R^{C\times H\times W}$ is rotated 90° counterclockwise along the W-axis. The rotated tensor is denoted as $z_2 \in R^{H\times C\times W}$. After passing through the Z-pool, the tensor $\hat{z}_2 \in R^{2\times C\times W}$ is obtained. Following convolutional layers and batch normalization layers, the output is $\hat{z}_2^* \in R^{1\times C\times W}$. The feature map then generates the attention weight w_2 through the sigmoid function. Finally, the output tensor $z_2 \cdot w_2$ is rotated 90° clockwise along the W-axis.

The two branches will get two feature maps, and then, the two feature maps will be aggregated together through a simple average calculation. Therefore, it can be expressed as follows:

$$y = \frac{1}{2}\big(R_1^*(R_1(z) \cdot \sigma(\psi_1(R_1(z))) + R_2^*(R_2(z) \cdot \sigma(\psi_2(R_2(z))) \tag{5}$$

where $R_i(\cdot)$ and $R_i^*(\cdot)$ represent the clockwise and counterclockwise rotation operations of the ith branch, respectively; ψ_i denotes the combination of Z-pool and convolution operation; and σ represents the sigmoid function.

The SSFIE module completes multidimensional interaction without reducing the dimensionality through cross-dimensional interaction, requiring minimal computing power and almost no additional parameters. By using the SSFIE module, the feature interaction between spectral–spatial of HS or MS data can be better realized and image features can be fully exploited. This article adds the SSFIE module behind the 3-D feature extraction network, allowing the network to automatically learn cross-dimensional interactions, enhance the weight of important feature channels, and suppress irrelevant information. As a result, the model can focus on more discriminative features, thereby improving the classification effect. Subsequently, an FSL method based on adaptive joint similarity metric learning is proposed for classification.

C. Few-Shot Learning

1) Adaptive Joint Similarity Metric: The widely used metric in existing FSL methods is Euclidean distance or cosine similarity, and few methods explore the fusion of metrics. Inspired by BSNet [30], this article proposes an adaptive joint similarity metric criterion to evaluate the source domain and target domain FSL. Specifically, given an FSL task T, C_k represents the sample set of class k. The prototype of the kth class is expressed as follows:

$$c_k = \frac{1}{C_k} \sum_{(x_i, y_i)\in C_k} f(x_i) \tag{6}$$

where $f(x_i)$ represents the feature extraction module. The adaptive similarity measure for the validation set samples x_j and prototype samples can be expressed as

$$\text{adm}(f(x_j), c_k) = \alpha \cdot \text{euc}(f(x_j), c_k) + \beta \cos(f(x_j), c_k). \tag{7}$$

Among them, α and β denote the weights that need to be learned and they are updated with cross-entropy loss. $\text{adm}(\cdot)$ represents the adaptive joint similarity metric, $\text{euc}(\cdot)$ corresponds to the Euclidean distance, and $\cos(\cdot)$ represents the cosine distance.

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2) FSL in the Source Domain and Target Domain: Whether performing FSL in the source domain or the target domain, N classes of samples are first randomly selected from the data to form an episode. Then, M samples from each of the selected N classes are used as the support set $S_s = \{(x_i, Y_i)\}_{i=1}^{N\times M}$, and K samples from the same class are selected as the query set $Q_s = \{(x_j, Y_j)\}_{i=1}^{N\times K}$. The source domain prediction probability can be expressed as follows:

$$\Gamma\left(\hat{Y}_j = k \mid \hat{x}_j \in Q\right) = \frac{\exp(-\text{adm}(f(x_j), c_k))}{\sum_{i=1}^{N}\exp(-\text{adm}(f(x_j), c_k))}. \quad (8)$$

Among them, adm$(\cdot)$ represents the adaptive joint similarity metric, x denotes the samples in the query set, and N represents the number of classes in each episode. The FSL loss of the support set in the source domain can be expressed as

$$L_{\text{fsl}}^{s} = E_{S_s,Q_s}\left[-\sum_{(x,Y)\in Q_s}\log\Gamma_\theta(Y=k|x)\right] \quad (9)$$

$$L_{\text{fsl}}^{t} = E_{S_t,Q_t}\left[-\sum_{(x,Y)\in Q_t}\log\Gamma_\theta(Y=k|x)\right] \quad (10)$$

where Y denotes the true label, S_s and S_t are the support set of the two domains, and Q_s and Q_t are query sets of them.

D. Discriminative Joint Probability Adaptation

Most transfer learning metrics primarily focus on increasing transferability between domains but overlook the distinguishability between different classes, which may impact classification performance. This article employs a DJPA module [15] to align source and target domain data, thereby maximizing both domain transferability and class distinguishability. According to the Bayesian law, the joint distribution $\text{P}(X|Y)$ can be expressed by $\text{P}(X|Y)\cdot\text{P}(Y)$, so the difference in the joint probability distribution between the source and the target domain can be expressed as follows:

$$\begin{aligned}
&d(\mathcal{D}_s,\mathcal{D}_t)\\
&= d(\text{P}(X_s\mid Y_s)\text{P}(Y_s),\text{P}(X_t\mid Y_t)\text{P}(Y_t))\\
&= \sum_{c=\hat{c}}^{C}\sum_{\hat{c}=1}^{C} d\left(\text{P}\left(X_s\mid Y_s^c\right)\text{P}\left(Y_s^c\right),\text{P}\left(X_t\mid Y_t^{\hat{c}}\right)\text{P}\left(Y_t^{\hat{c}}\right)\right)\\
&\quad+\sum_{c\neq c}\sum_{\hat{c}=1}^{C} d\left(\text{P}\left(X_s\mid Y_s^c\right)\text{P}\left(Y_s^c\right),\text{P}\left(X_t\mid Y_t^{\hat{c}}\right)\text{P}\left(Y_t^{\hat{c}}\right)\right)\\
&= \sum_{c=1}^{C} d\left(\text{P}\left(X_s\mid Y_s^c\right)\text{P}\left(Y_s^c\right),\text{P}\left(X_t\mid Y_t^c\right)\text{P}\left(Y_t^c\right)\right)\\
&\quad+\sum_{c\neq c}\sum_{\hat{c}=1}^{C} d\left(\text{P}\left(X_s\mid Y_s^c\right)\text{P}\left(Y_s^c\right),\text{P}\left(X_t\mid Y_t^{\hat{c}}\right)\text{P}\left(Y_t^{\hat{c}}\right)\right)\\
&= \mathcal{M}_T+\mathcal{M}_D
\end{aligned} \quad (11)$$

where $\mathcal{M}_T$ denotes the probability difference of the same class between the two domains and $\mathcal{M}_D$ represents the probability difference for different classes of data between the two domains. DJPA not only aligns the distribution of data for the same class across domains but also increases the difference in the distribution of data for different classes across domains, thereby improving the discriminative information of cross-domain data. Consequently, the discriminative joint probability distribution difference is defined as follows:

$$d(\mathcal{D}_s,\mathcal{D}_t) = \mathcal{M}_T - u\mathcal{M}_D \quad (12)$$

$$\begin{aligned}
d(\mathcal{D}_s,\mathcal{D}_t) =& \sum_{c=1}^{C}\left\|\frac{1}{n_s}\sum_{i=1}^{n_s^c}A^\top\mathbf{x}_{s,i}^c-\frac{1}{n_t}\sum_{j=1}^{n_t^c}A^\top\mathbf{x}_{t,j}^c\right\|_2^2\\
&-u\sum_{c\neq c}\sum_{c=1}^{C}\left\|\frac{1}{n_s}\sum_{i=1}^{n_s^c}A^\top\mathbf{x}_{s,i}^c-\frac{1}{n_t}\sum_{j=1}^{n_t^{\hat{c}}}A^\top\mathbf{x}_{t,j}^{\hat{c}}\right\|_2^2
\end{aligned} \quad (13)$$

where $d(\mathcal{D}_s,\mathcal{D}_t)$ denotes the difference between the two domains; A is the mapping matrix; and n_s and n_t are the number of samples in the source and target domains, respectively.

E. Batch Nuclear-Norm Maximization

In order to improve the discriminability and diversity of the model, inspired by [42], the BNM module is introduced. This module utilizes the Frobenius norm to constrain the discriminability of the model and employs the rank of the matrix to constrain the diversity of the model.

Assume that the prediction matrix of the model for batch data is $Y\in\mathbb{R}^{B\times C}$, where B and C represent the batch size and the number of classes, respectively. The matrix $Y\in\mathbb{R}^{B\times C}$ satisfies the following conditions:

$$\begin{aligned}
&\sum_{j=1}^{C}Y_{ij}=1 \quad \forall i\in 1,\ldots,B\\
&Y_{ij}\geq 0 \quad \forall i\in 1,\ldots,B,\quad j=1,\ldots,C.
\end{aligned} \quad (14)$$

To constrain the discriminability of the model, the maximum $\|Y\|_F$ can be expressed as follows:

$$\|Y\|_F=\sqrt{\sum_{i=1}^{B}\sum_{j=1}^{C}\left|Y_{ij}\right|^2}. \quad (15)$$

By constraining the rank of the matrix Y to the maximum, the model can be prompted to treat different classes of data in a more balanced manner during the training process, thereby improving the diversity and generalization ability of the model. This approach helps prevent the model from overfocusing on certain classes during training, thereby improving the classification accuracy. Since optimizing the matrix rank is an NP-hard problem, the nuclear norm $\|Y\|_*$ is utilized as a convex approximation to constrain the rank. Therefore, the relationship between $\|Y\|_*$ and $\|Y\|_F$ can be expressed as

$$\frac{1}{\sqrt{D}}\|Y\|_*\leq\|Y\|_F\leq\|Y\|_*\leq\sqrt{D}\|Y\|_F \quad (16)$$

where $D=\min(B,C)$ indicates that $\|Y\|_*$ and $\|Y\|_F$ can restrict each other. We can then express this relationship as

$$\|Y\|_*\leq\sqrt{D}\|Y\|_F\leq\sqrt{DB}. \quad (17)$$

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By maximizing the constraint matrix $\|Y\|_*$, $\|Y\|_F$ can be driven to its maximum, thereby improving the discriminability and diversity of the model. Therefore, this component of the loss function can be expressed as follows:

$$L_{\text{bnm}} = -\left\|G(X^{\text{T}})\right\|_* \tag{18}$$

where X^{T} represents the target domain data and G stands for the deep network described in this article.

To summarize, the overall loss of the proposed method can be expressed as follows:

$$L = L_{\text{fsl}}^s + L_{\text{fsl}}^t + L_{\text{cd}} + L_{\text{bnm}} \tag{19}$$

where L_{fsl}^s and L_{fsl}^t are the losses of FSL in the source and target domains, respectively; L_{cd} is the loss of DJPA; and L_{bnm} is the loss of BNM.

F. Open-Set Classification Module

During the testing phase, the test data are first mapped to the same dimensions as the source domain data using the mapping layer. Subsequently, the trained deep residual network and SSFIE module are employed to extract discriminative features. Inspired by [31], the open-set nearest neighbor algorithm is used to classify the test data. Different from the traditional nearest neighbor method, the metric of the nearest neighbor algorithm in this article uses the joint metric. During the training phase of the open-set nearest neighbor algorithm, only the training samples need to be stored. In the testing phase, first obtain the two nearest test samples of the sample u to be tested, denoted as m and n. If these two nearest samples have the same label $\psi(m) = \psi(n)$, then the test sample has the same label as m and n. If the labels are different $\psi(m) \neq \psi(n)$, the calculation proceeds as follows:

$$R = \frac{\text{adm}(u, m)}{\text{adm}(u, n)}. \tag{20}$$

Among them, R represents the similarity measure between u and m in the feature space. If the value of R is less than or equal to the threshold T, then the label of u is consistent with m; otherwise, it is classified as an unknown class

$$\psi(u) = \begin{cases} \psi(t), & R \leq T \\ \text{Unk}, & R > T. \end{cases} \tag{21}$$

Through the open-set classification module, the known classes and unknown classes of MS features in the target domain can be distinguished.

III. Experimental Results and Discussion

To validate the effectiveness of the proposed method, experiments were conducted on four HS–MS datasets.

A. Data Description

1) University of Houston: The University of Houston (UH) dataset covers HS images of the Houston area as well as MS images generated through simulations. The HS image comes from the ITRES CASI-1500 sensor and was provided by the 2013 IEEE GRSS data fusion competition. The spatial resolution of the image is 2.5 m, and its size is 349 × 1905 pixels. It covers the spectral range from 364 to 1046 nm, with a total of 144 bands. The MS image simulates the Sentinel-2 MS image by downsampling the HS data using the spectral response function, resulting in only ten bands. For source domain HS data, this article uses 50 fixed samples in the overlapping area of the HS dataset for training. For the target domain data, this article randomly selected ten samples in the overlapping area of each class to participate in the training process. It is worth noting that unknown classes are not included when training the model. During testing, the model can not only classify samples of known classes but also distinguish unknown classes in MS images. The false-color image of the HS–MS data of the UH dataset is shown in Fig. 4, and Table I presents the specific information regarding data classes and quantities.

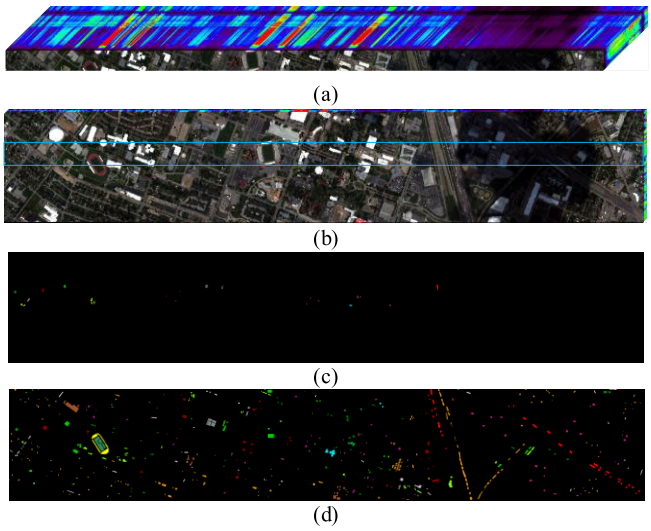


Fig. 4. UH dataset. (a) HS false-color composite image, (b) MS false-color composite image, (c) training labels, and (d) test labels.

TABLE I
Information for the UH Dataset

Class No.	Color	Class Name	TR in Source	TE
1		Healthy Grass	50	1186
2		Stressed Grass	50	1165
3		Synthetic Grass	50	647
4		Tree	50	1194
5		Soil	50	1192
6		Water	30	295
7		Commercial	50	1194
8		Road	50	1175
9		Highway	50	1177
10		Parking Lot 1	50	1183
11		Tennis Court	50	378
12		Running Track	50	610
13		Unknown	-	2822
Total			580	14218

2) Yellow River Estuary_1: The Yellow River Estuary_1 (YRE_1) dataset includes HS images captured by the GF5 satellite, along with a simulated MS image. The MS image is derived by downsampling the GF5 HS image using the GF1 spectral response function, resulting in four bands covering a

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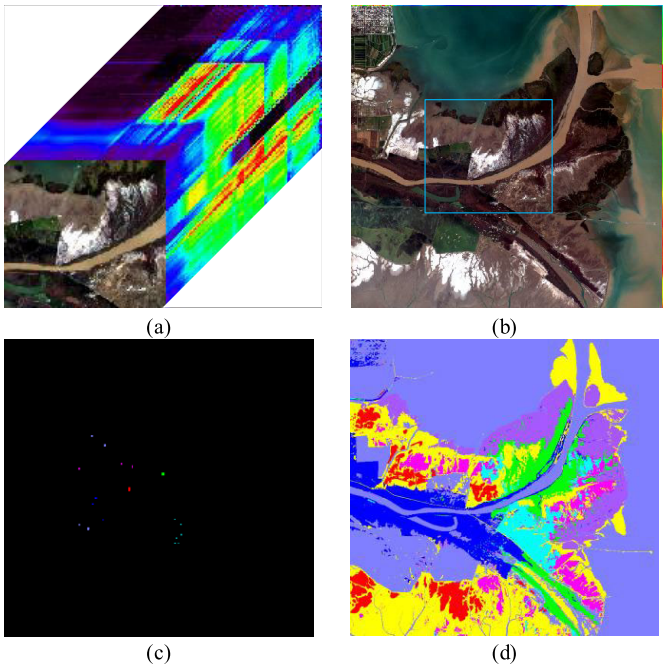


Fig. 5. YRE_1 dataset. (a) HS false-color composite image, (b) MS false-color composite image, (c) training labels, and (d) test labels.

TABLE II
INFORMATION FOR THE YRE_1 AND YRE_2 DATASETS

Class No.	Color	Class Name	TR in Source	TE YRE_1	TE YRE_2
1		Reed	50	46555	98275
2		Tamarix	50	26109	55139
3		Tidal Flat Reed	50	20667	43891
4		Salt Marsh	50	18952	40126
5		Suaeda Salsa	50	25477	53874
6		Water	50	266686	562243
7		Spartina Alterniflora	50	44882	94479
8		Unknown	-	110314	232701
Total			350	559642	1180728

spectral range of 450–890 nm. The HS image was captured in the Yellow River Estuary area on November 1, 2018, spanning a spectral range of 390–2513 nm with a total of 330 bands. The spatial resolution for both the HS and MS data is 30 m. Similar to the approach used with the UH dataset, the source domain selects 50 fixed samples from the overlapping area for training, while the target domain randomly selects ten training samples. Fig. 5 presents the false-color map of the YRE_1 dataset, which includes a total of eight classes. Specific information regarding these classes is provided in Table II.

3) Yellow River Estuary_2: The Yellow River Estuary_2 (YRE_2) dataset consists of GF5 HS data and WFV-GF1 Yellow River Estuary data. The GF5 data are identical to that of the YRE_1 dataset, which was captured on November 1, 2018. The HS image was upsampled to match the spatial resolution of the WFV-GF1 data using nearest neighbor interpolation. The WFV-GF1 data were captured in the Yellow River Estuary area on October 31, 2018. It has a size of 1066 × 1108 pixels and a spatial resolution of 16 m, with a total of four bands. The training labels, test labels, and false-color images of the YRE_2 dataset are similar to those in the YRE_1 dataset. Detailed settings for the training and testing samples are provided in Table II.

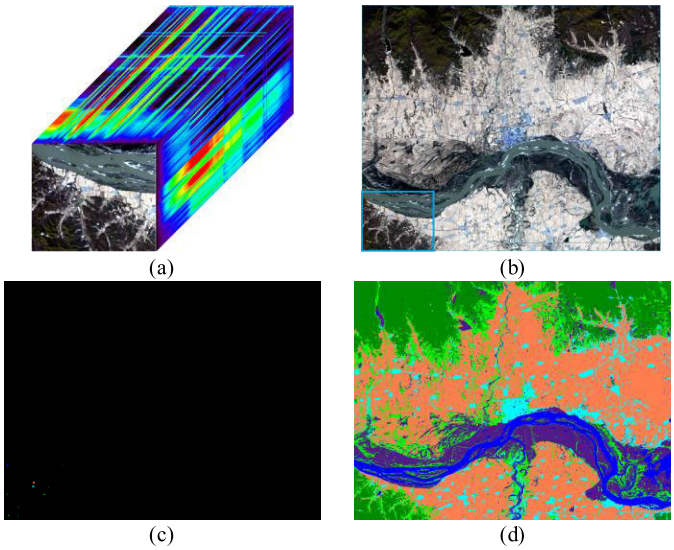


Fig. 6. Harbin dataset. (a) HS false-color composite image, (b) MS false-color composite image, (c) training labels, and (d) test labels.

TABLE III
INFORMATION FOR THE HARBIN DATASET

Class No.	Color	Class Name	TR in Source	TE
1		Trees	50	215919
2		Water	50	73539
3		Farmland	50	528098
4		Towns	50	95351
5		Scrub	50	160410
6		Unknown	-	167094
Total			250	1240411

4) Harbin: The Harbin dataset is composed of GF5 HS data and WFV-GF1 data. The HS image was captured in the Harbin area on October 16, 2019, covering a spectral range from 390 to 2513 nm with a total of 330 bands. After removing the bad bands, 305 bands remain in the HS image, with a spatial resolution of 30 m. Similar to the YRE_1 dataset, the HS image was upsampled to match the spatial resolution of the MS image. The MS image was taken in the Harbin area on October 17, 2019, with a spatial resolution of 16 m, a total of four bands, and a size of 967 × 1283. Fig. 6 shows the false-color image of the Harbin dataset. The known classes and unknown classes contain a total of six classes. Table III gives the specific class information.

B. Experimental Setting

To evaluate the effectiveness of our proposed method, our method is compared with five related methods, namely, OSNN [31], open-set classification-supervised contrastive learning (OSC-SCL) [33], ProtoNet [22], FEAT [24], and SnaTCHer [40]. Among them, OSNN and OSC-SCL only involve one domain. OSNN is an open-set classification method that calculates the ratio of similarity scores between the two most similar classes in a sample. It then compares this ratio with a preset threshold to distinguish unknown classes.

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TABLE IV
PA, OS, OS*, HOS, OA, AND KAPPA FOR THE UH DATASET

Class \ Method	OSNN	OSC-SCL-H	OSC-SCL-M	ProtoNet	FEAT	SnaTCHer	Ours
Healthy Grass	73.30	80.09	71.31	87.62	75.32	79.93	**92.88**
Stressed Grass	88.14	**98.68**	81.64	69.62	81.48	85.43	82.55
Synthetic Grass	**99.85**	97.56	98.56	92.53	87.66	98.51	98.57
Tree	78.21	84.88	**97.02**	46.14	72.82	85.80	75.09
Soil	88.86	89.53	95.75	70.12	76.65	95.87	**96.54**
Water	90.15	87.07	82.76	66.15	81.53	85.53	**91.69**
Commercial	32.83	72.91	65.67	**74.19**	68.40	47.89	64.56
Road	71.83	**81.79**	78.44	53.65	41.22	75.26	73.22
Highway	44.98	56.25	33.01	46.69	63.48	**91.18**	69.19
Parking Lot1	14.43	49.79	14.84	**87.02**	76.80	56.20	72.75
Tennis Court	99.32	99.42	99.27	85.28	88.78	99.29	**99.43**
Running Track	95.30	**99.54**	96.96	98.84	96.06	87.27	92.42
UNK	7.81	45.01	26.58	4.73	6.35	38.10	**60.43**
OS	68.08	80.20	72.45	67.89	70.51	78.95	**82.26**
OS*	73.10	83.13	76.27	73.16	75.85	82.35	**84.07**
HOS	14.13	58.39	39.42	8.87	11.73	52.10	**70.32**
OA	55.65	73.30	63.23	56.86	59.91	71.68	**77.39**
Kappa	0.5188	0.7044	0.5966	0.5306	0.5632	0.6884	**0.7474**

TABLE V
PA, OS, OS*, HOS, OA, AND KAPPA FOR THE YRE_1 DATASET

Class \ Method	OSNN	OSC-SCL-H	OSC-SCL-M	ProtoNet	FEAT	SnaTCHer	Ours
Reed	33.04	**57.68**	45.36	12.86	32.53	22.68	47.37
Tamarix	56.87	40.13	23.02	**79.25**	40.94	20.56	58.21
Tidal Flat Reed	21.11	64.14	15.37	43.61	**78.01**	40.28	45.93
Salt Marsh	95.25	**97.04**	95.73	96.32	95.72	74.47	71.69
Suaeda Salsa	**83.50**	40.22	64.80	76.44	57.84	63.48	61.75
Water	78.15	89.92	81.02	77.43	79.98	85.51	**91.71**
Spartina Altemiflora	76.69	73.59	75.57	81.14	84.22	71.14	**87.63**
UNK	2.88	29.47	39.05	8.80	11.64	53.10	**70.54**
OS	55.94	61.52	54.99	59.48	60.12	53.90	**66.85**
OS*	63.52	66.10	57.27	66.72	**67.04**	54.02	66.32
HOS	5.51	40.76	46.44	15.56	19.84	53.56	**68.36**
OA	57.17	70.05	63.97	57.86	60.53	66.65	**72.87**
Kappa	0.4523	0.5936	0.5165	0.4562	0.4909	0.5424	**0.6263**

TABLE VI
PA, OS, OS*, HOS, OA, AND KAPPA FOR THE YRE_2 DATASET

Class \ Method	OSNN	OSC-SCL-H	OSC-SCL-M	ProtoNet	FEAT	SnaTCHer	Ours
Reed	25.80	**57.67**	17.80	8.74	14.96	37.64	55.73
Tamarix	32.81	40.13	34.38	**70.73**	63.28	34.21	43.98
Tidal Flat Reed	30.06	**64.14**	62.47	20.86	45.57	29.91	54.94
Salt Marsh	94.51	**97.04**	90.64	94.29	98.21	77.46	87.69
Suaeda Salsa	**62.31**	40.22	35.45	41.32	39.33	59.55	47.39
Water	75.26	89.92	88.42	77.69	77.07	82.56	**90.57**
Spartina Altemiflora	80.25	73.59	81.36	85.63	84.96	70.98	**86.38**
UNK	15.55	29.47	26.48	16.23	17.21	50.46	**67.37**
OS	52.07	61.52	54.63	51.94	55.08	55.35	**66.75**
OS*	57.29	66.10	58.65	57.04	60.49	56.05	**66.67**
HOS	24.47	40.76	36.49	25.27	26.79	53.12	**67.02**
OA	56.17	70.05	63.94	56.71	57.92	66.13	**72.35**
Kappa	0.4384	0.5936	0.5085	0.441	0.4591	0.5398	**0.6207**

OSC-SCL is an open-set HS image classification method that constrains spectral and spatial features through supervised contrastive learning and identifies unknown classes using the class anchor strategy. The OSC-SCL-H and OSC-SCL-M are used as baseline methods. OSC-SCL-H is trained and tested with HS data, while OSC-SCL-M employs MS data for both training and testing. We expect that the accuracy of our proposed method surpasses that of OSC-SCL-M and either exceeds or closely matches OSC-SCL-H. This would validate that the inclusion of HS data can improve the classification ability of MS images. The ProtoNet method and the FEAT method are closed-set few-shot classification methods. The ProtoNet method learns the embedding space and identifies the prototype center of each class to represent the class, which can

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TABLE VII
PA, OS, OS*, HOS, OA, AND KAPPA FOR THE HARBIN DATASET

Class \ Method	OSNN	OSC-SCL-H	OSC-SCL-M	ProtoNet	FEAT	SnaTCHer	Ours
Trees	90.36	88.33	90.11	75.83	89.18	77.79	**92.68**
Water	84.29	**91.03**	86.43	81.47	88.68	69.31	78.05
Farmland	68.73	83.90	71.38	80.84	79.24	78.56	**87.75**
Towns	31.15	47.90	**58.39**	41.12	45.38	33.14	39.62
Scrub	50.11	37.01	28.59	57.16	38.12	**64.42**	43.71
UNK	7.97	36.91	29.24	3.42	6.592	34.67	**60.44**
OS	55.44	64.19	60.81	56.64	57.86	59.65	**67.04**
OS*	64.93	**69.64**	67.11	67.28	68.12	64.65	68.36
HOS	14.2	48.26	40.74	6.52	12.02	45.14	**63.57**
OA	59.57	69.94	63.44	62.455	63.40	66.64	**73.15**
Kappa	0.4784	0.6039	0.5293	0.5113	0.5215	0.5632	**0.6416**

better utilize the similarity information between samples compared with traditional feature-based methods. FEAT introduces a task-adaptive feature transformation method that makes instance embeddings task-specific through set-to-set transformation, effectively enhancing classification performance under conditions of limited samples. As in the implementation process of [40], we use the negative value of the predicted class probability to determine the unknown detection score, thereby distinguishing the unknown class. Both the ProtoNet method and the FEAT method utilize meta-training, which trains on the HS image and tests on the MS image. SnaTCHer is an open-set cross-domain few-shot classification method. It measures the difference between the transformed prototype and the modified prototype set based on transformation consistency, thereby transforming the problem of estimating unknown class distribution into a relative feature conversion problem. For all few-shot classification methods, 50 samples from the source domain are selected for training. We randomly select ten labeled samples from each class in the target domain as training samples and the others as test samples. We strive to ensure fairness in our experiments by conducting each set ten times to guarantee result reliability and calculating the average accuracy metrics.

In the proposed method, we refer to the settings of [25], [26], and [29], which utilize a 9×9 neighborhood as the spatial size of the input HS and MS images. The convolution kernel is initialized with Xavier normalization and trained using the Adam optimizer. To enhance the learning of each class prototype, the number of iterations is set to 10 000, with a learning rate of 0.001 for network convergence. Each episode involves an N-way, K-shot classification task, where N represents the number of classes in the target domain, determined by the datasets used in the experiment. K is the number of labeled samples randomly selected from each class, and K is set to 1 in the experiment. The number of samples of each class in the query set is denoted as Nq. Liu et al. [25], Li et al. [26], and Zhang et al. [29] indicate that augmenting the number of query samples enhances the simulation of few-shot classes in the target domain. In this experiment, Nq was set to 19. During the testing phase, discriminative features of the target are extracted using the trained feature extractor. Subsequently, the unknown class classification strategy developed in this article is employed to accurately classify known classes while also identifying unknown classes. This article employs overall accuracy (OA), producer accuracy (PA), and Kappa coefficient (kappa) to evaluate the classification accuracy. The classification accuracy of unknown classes is denoted by UNK. For evaluating open-set classification accuracy, the same evaluation metrics are utilized as previous work [32], [38], including known class average open-set accuracy (OS), and the harmonic mean of OS* and UNK (HOS). OS includes unknown classes, representing the average per-class accuracy across all classes. OS* represents the average per-class accuracy for known classes. HOS provides a balanced assessment of recognition performance for both known and unknown classes. These indicators can be expressed as follows:

$$\mathrm{UNK} = \frac{\left|x : x \in \mathcal{D}_t^{\mathrm{unk}} \wedge \hat{y}(x) = \mathrm{unk}\right|}{\left|x : x \in \mathcal{D}_t^{\mathrm{unk}}\right|} \tag{22}$$

$$\mathrm{OS}^* = \frac{1}{C}\sum_{i=1}^{C} \frac{\left|x : x \in \mathcal{D}_t^{i} \wedge \hat{y}(x) = i\right|}{\left|x : x \in \mathcal{D}_t^{i}\right|} \tag{23}$$

$$\mathrm{OS} = \frac{1}{C+1}\sum_{i=1}^{C+1} \frac{\left|x : x \in \mathcal{D}_t^{i} \wedge \hat{y}(x) = i\right|}{\left|x : x \in \mathcal{D}_t^{i}\right|} \tag{24}$$

$$\mathrm{HOS} = \frac{2 \times \mathrm{OS}^* \times \mathrm{UNK}}{\mathrm{OS}^* + \mathrm{UNK}}. \tag{25}$$

C. Experimental Results and Analysis

The comparative experimental results of the proposed method and the five related methods are displayed in Tables IV–VII. The best results of the experiment are highlighted in bold. The effectiveness of the proposed method surpasses that of the comparative methods, even outperforming the baseline method OSC-SCL-H, thereby validating its efficacy. The performance of the proposed method and the comparison methods on the four datasets is introduced and analyzed in detail next.

Table IV presents the classification results of the UH dataset. It is apparent from the table that our method excels across all indicators. Our method achieves an OA of 77.39%, which is 5.71% higher than the second-highest performing method, SnaTCHer. For known classes, our method achieves the best classification results across four classes, and it also achieves

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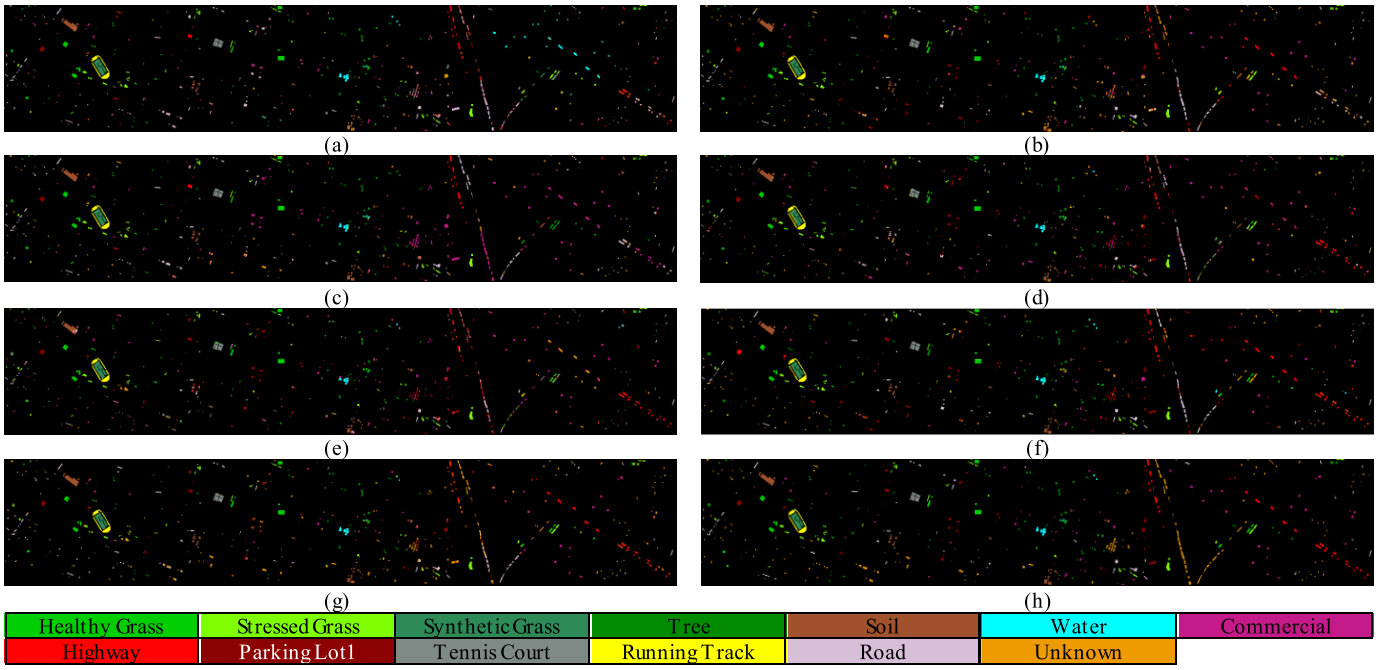


Fig. 7. Classification results on the UH dataset. (a) OSNN, (b) OSC-SCL-H, (c) OSC-SCL-M, (d) ProtoNet, (e) FEAT, (f) SnaTCHer, (g) Ours, and (h) ground truth.

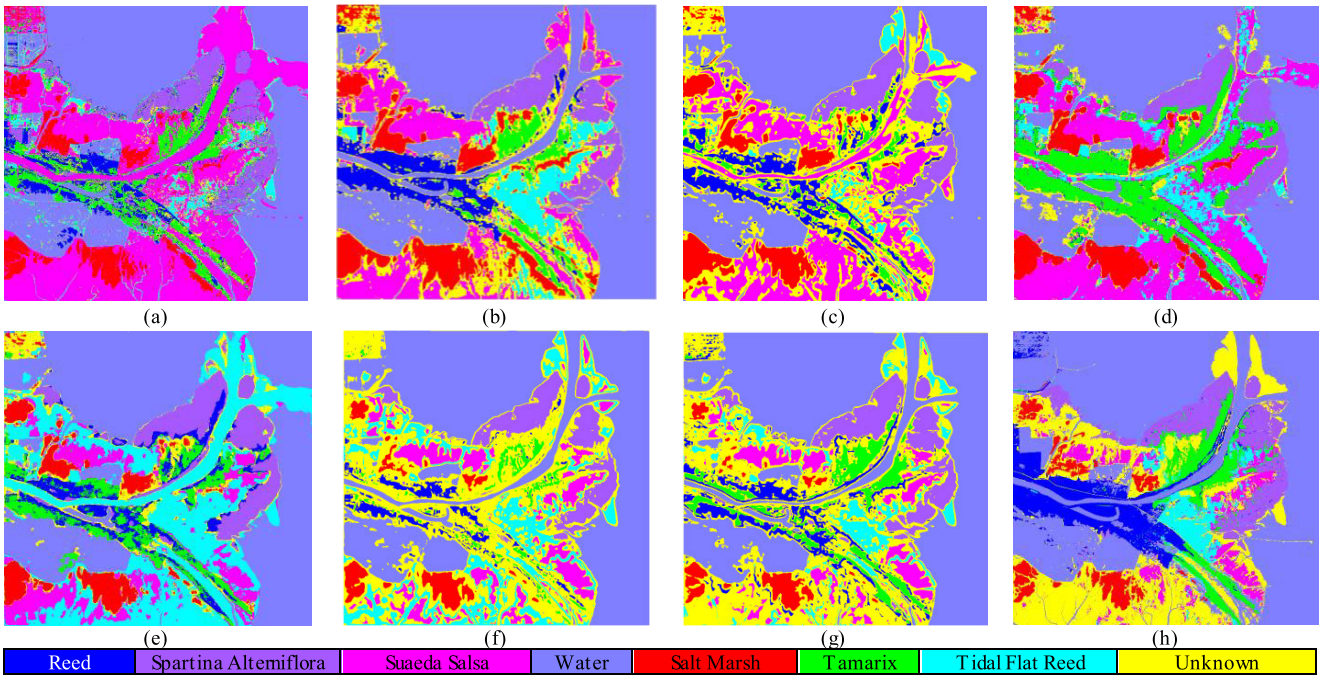


Fig. 8. Classification results on the YRE_1 dataset. (a) OSNN, (b) OSC-SCL-H, (c) OSC-SCL-M, (d) ProtoNet, (e) FEAT, (f) SnaTCHer, (g) Ours, and (h) ground truth.

TABLE VIII
SET ARCHITECTURES WITH DIFFERENT MODULES

	FSL	SSFIE	DJPA	BNM
Ours-V0	√			
Ours-V1		√	√	√
Ours-V2	√	√	√	
Ours-V3	√	√		√
Ours-V4	√		√	√
Ours	√	√	√	√

optimal results for unknown classes. The OSNN method identifies unknown classes by considering the distance ratio between the target point and surrounding points. However, the OSNN method demonstrates the poorest classification performance, with an OA of only 55.65% and an HOS of 14.13%. This is because OSNN is an open-set classification method, which is less suitable for scenarios with limited samples. In such cases, neither known nor unknown classes can achieve satisfactory classification results. The OSC-SCL method utilizes contrastive supervised learning to aggregate known classes and separate the unknown classes. This approach enables open-set classification even in scenarios with limited labeled samples. Comparatively, the classification results of

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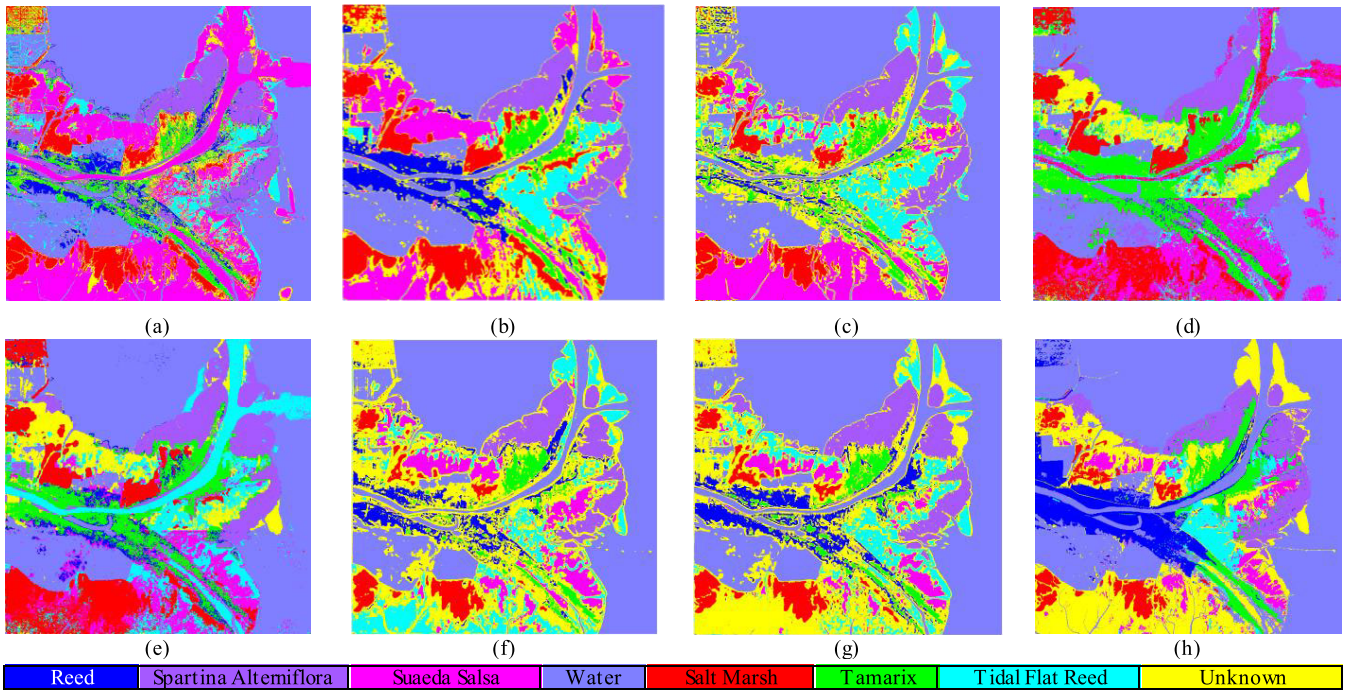


Fig. 9. Classification results on the YRE_2 dataset. (a) OSNN, (b) OSC-SCL-H, (c) OSC-SCL-M, (d) ProtoNet, (e) FEAT, (f) SnaTCHer, (g) Ours, and (h) ground truth.

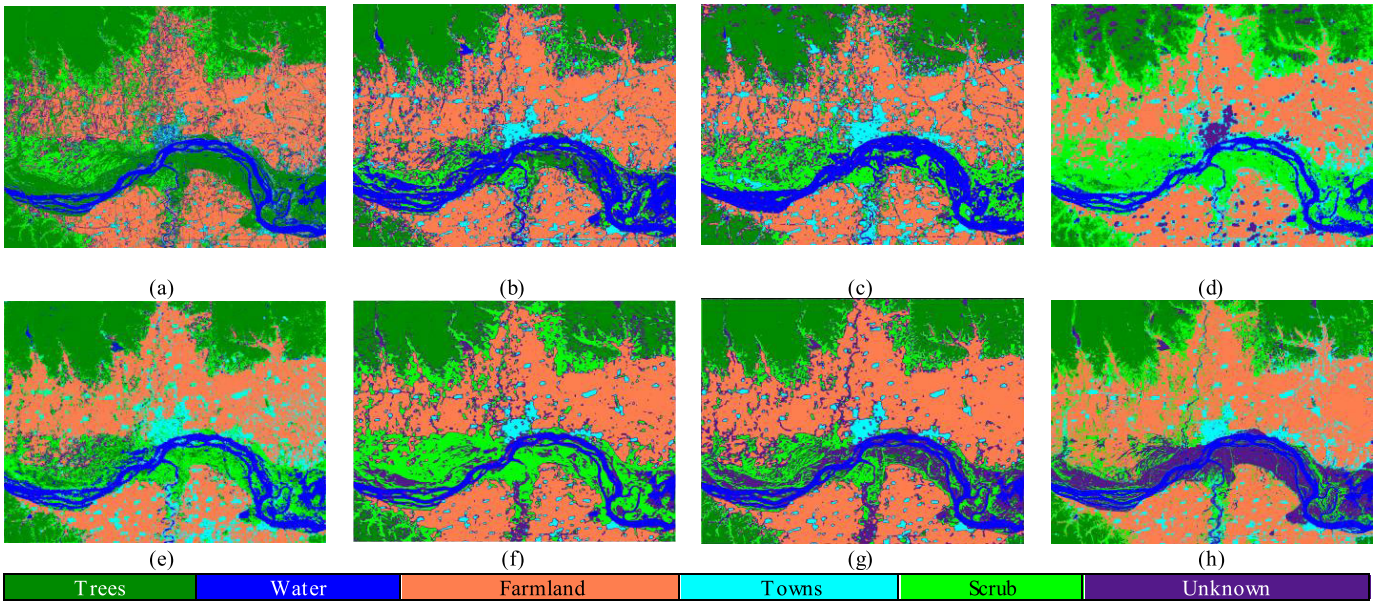


Fig. 10. Classification results on the Harbin dataset. (a) OSNN, (b) OSC-SCL-H, (c) OSC-SCL-M, (d) ProtoNet, (e) FEAT, (f) SnaTCHer, (g) Ours, and (h) ground truth.

OSC-SCL-H outperform OSC-SCL-M, indicating that HS images contain richer spectral information than MS images. Despite the introduction of HS images to assist MS image classification, ProtoNet and FEAT achieve lower classification results than OSC-SCL-M. Remarkably, the HOS of the ProtoNet method is the lowest among all methods, indicating its inferior recognition performance on unknown classes and the misclassification of most unknown classes into known classes. The ProtoNet method represents each class by identifying the prototype center within the embedding space and then calculates the distance between the test sample and each class prototype to perform classification. However, it relies solely on the Euclidean distance as the loss function, which overlooks the potential impact of the spatial distribution of sample representations. This approach often maps unknown class samples to known classes, leading to poor classification results for unknown classes. The FEAT method introduces a task-adaptive feature transformation module based on class prototypes, allowing the model to learn discriminative features tailored to specific object classification tasks. However, since the method is designed for closed-set few-shot classification, it overlooks the differences between known classes and unknown samples,

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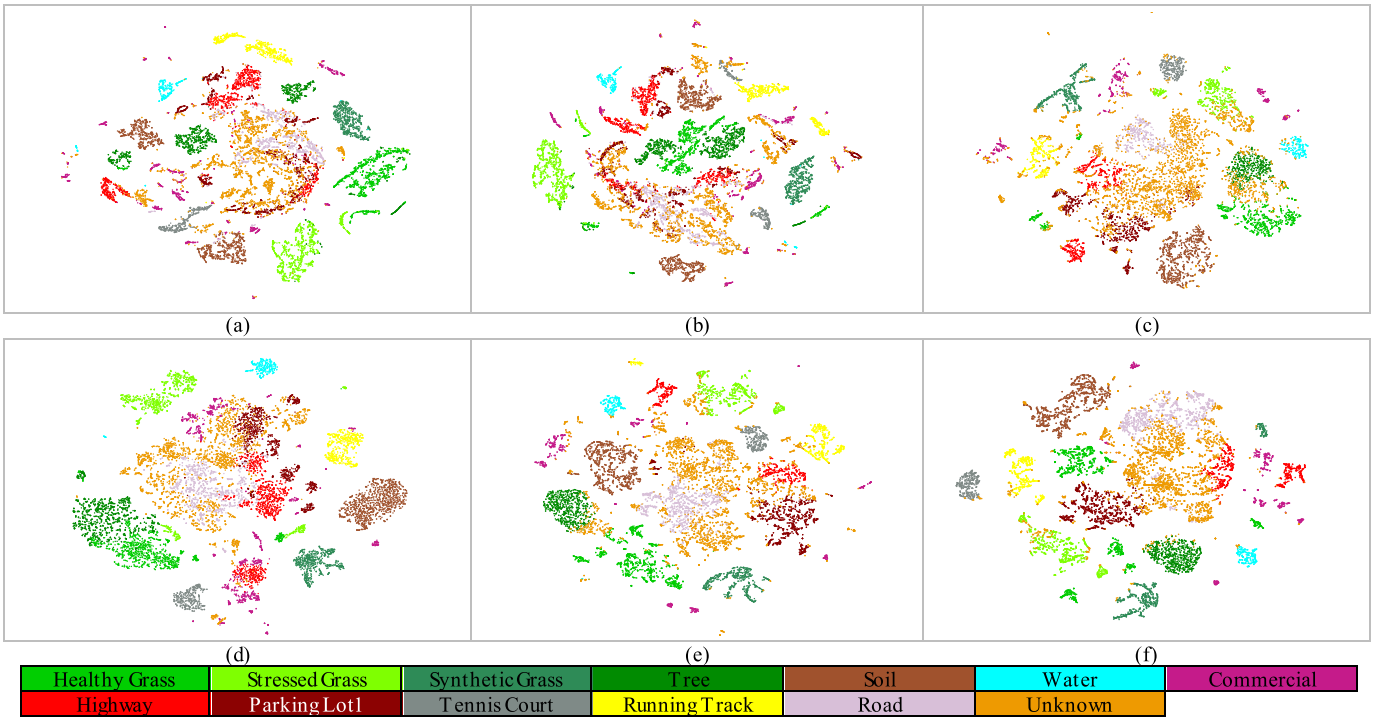


Fig. 11. t-SNE visualization on the Houston dataset. Features of (a) MS image, (b) HS image, (c) ProtoNet, (d) FEAT, (e) SnaTCHer, and (f) Ours method.

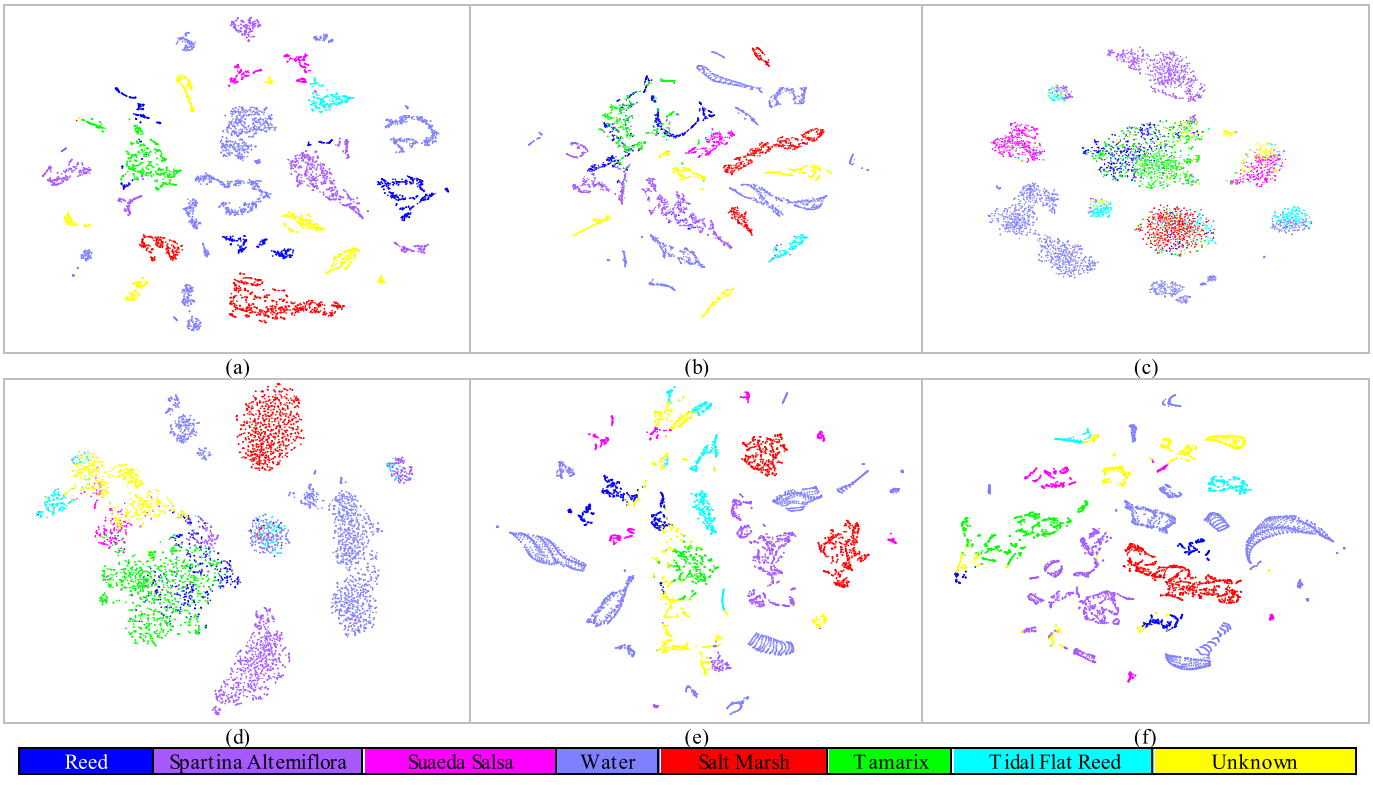


Fig. 12. t-SNE visualization on the YRE_1 dataset. Features of (a) MS image, (b) HS image, (c) ProtoNet, (d) FEAT, (e) SnaTCHer, and (f) Ours method.

resulting in suboptimal performance for unknown classes. The SnaTCHer method combines FSL and open-set classification to recognize unknown classes by training task-adaptive feature transformers, achieving the second-highest effectiveness. The method proposed in this article achieves the best results, indicating that feature interaction enhancement and adaptive measurement can help the model make more accurate decisions. In order to compare the classification effects more intuitively, Fig. 7 shows the classification results of these methods. It can be seen from the figure that the ProtoNet method and the FEAT method mistakenly classify most unknown classes into the Road class. Our method is

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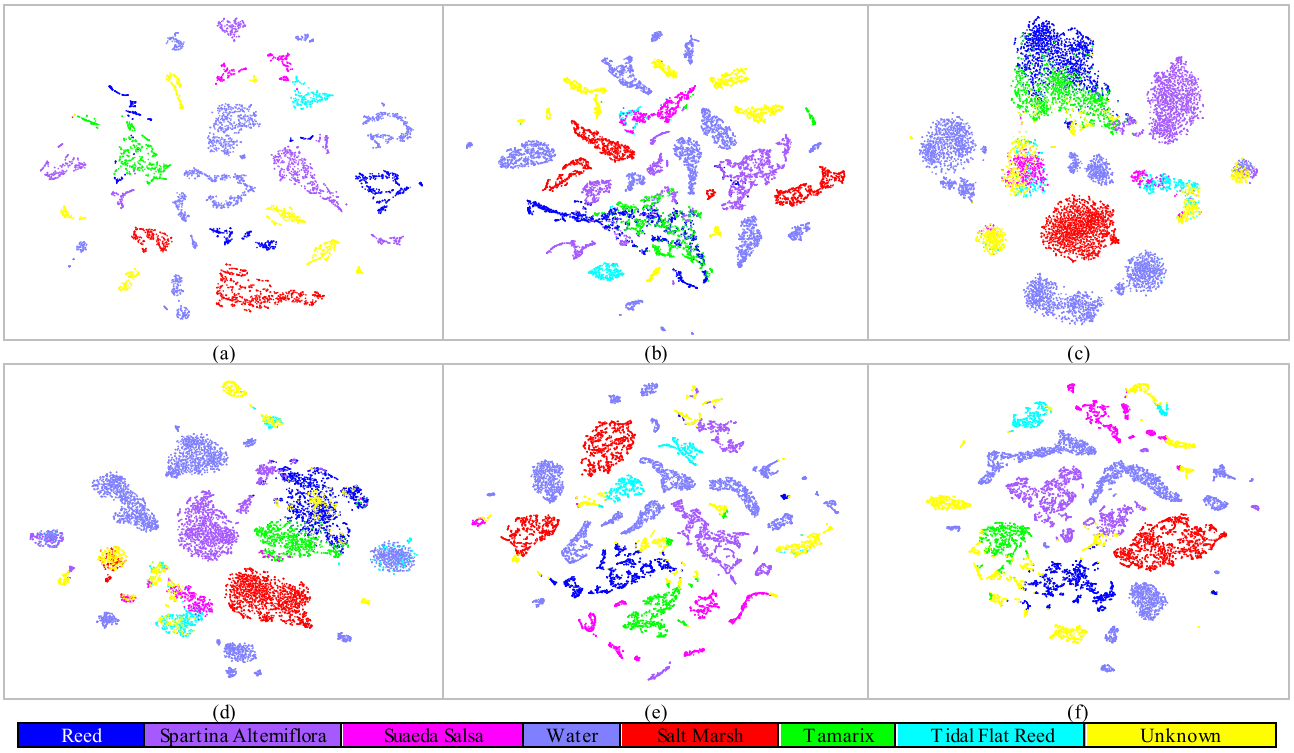


Fig. 13. t-SNE visualization on the YRE_2 dataset. Features of (a) MS image, (b) HS image, (c) ProtoNet, (d) FEAT, (e) SnaTCHer, and (f) Ours method.

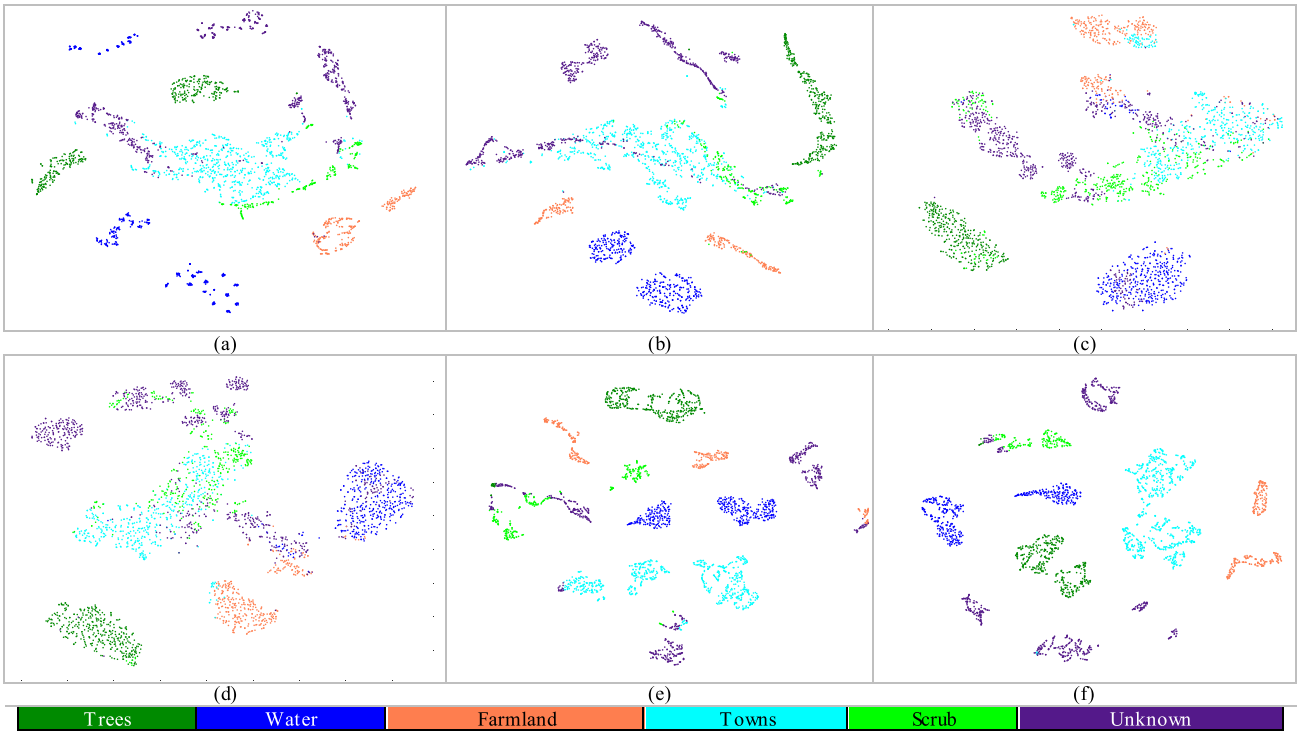


Fig. 14. t-SNE visualization on the Harbin dataset. Features of (a) MS image, (b) HS image, (c) ProtoNet, (d) FEAT, (e) SnaTCHer, and (f) Ours method.

closest to the ground truth and achieves the best classification results.

As depicted in Table V, the classification results of the YRE_1 dataset are presented. The results show that our method surpasses the other methods in the evaluation metrics, including OS, HOS, OA, and Kappa. Due to the superior spectral resolution capability of HS imagery, the classification results of OSC-SCL-H are superior to those of OSC-SCL-M. OSNN exhibits the poorest classification results, with all FSL classification methods outperforming OSNN. This demonstrates that under conditions of limited samples, FSL can acquire a more effective metric space, thereby enhancing the

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TABLE IX
ABLATION COMPARISON OF EACH VARIANT

	Ours-V0	Ours-V1	Ours-V2	Ours-V3	Ours-V4	Ours
Dataset	University of Houston					
OA	71.34	58.93	76.51	75.68	74.32	77.39
	±2.14	±2.08	±1.25	±1.62	±1.53	±1.16
Kappa	68.53	44.72	73.49	71.84	71.46	74.74
	±2.35	±2.21	±1.37	±1.83	±1.72	±1.31
HOS	53.67	47.05	61.97	62.07	63.74	70.32
	±2.53	±2.41	1.56	±1.96	±1.85	±1.52
Dataset	YELLOW RIVER ESTUARY 1					
OA	67.93	54.56	71.74	70.36	70.13	72.87
	±3.24	±2.64	±1.65	±1.93	±1.73	±1.52
Kappa	54.32	40.87	60.82	59.61	59.39	62.63
	±3.54	±2.81	±1.83	±2.13	±1.82	±1.71
HOS	47.81	36.83	55.73	58.97	56.52	68.36
	±3.83	±2.97	±2.01	±2.29	±1.96	±1.86
Dataset	YELLOW RIVER ESTUARY_2					
OA	67.26	53.75	71.24	70.02	69.23	72.35
	±3.37	±2.83	±1.74	±2.05	±1.81	±1.65
Kappa	54.01	39.04	60.45	59.16	58.27	62.07
	±3.26	±2.96	±1.89	±2.21	±2.01	±1.87
HOS	47.41	36.05	55.32	58.41	56.41	67.02
	±3.97	±3.17	±2.08	±2.37	±2.24	±2.06
Dataset	HARBIN					
OA	65.47	50.62	71.03	69.86	67.52	73.15
	±2.11	±2.04	±1.31	±1.61	±1.58	±1.21
Kappa	53.43	42.41	60.98	60.21	57.36	64.16
	±2.37	±2.26	±1.48	±1.79	±1.74	±1.43
HOS	46.53	40.32	53.82	55.69	54.43	63.57
	±2.61	±2.53	±1.63	±2.01	±1.90	±1.62

classification efficacy. The closed-set few-shot classification methods ProtoNet and FEAT have poor classification results for the unknown class. This is primarily because ProtoNet and FEAT are closed-set few-shot classification methods designed for closed-set classification tasks, resulting in limited capabilities for classifying unknown instances. The few-shot open-set classification method Snatcher is better than the ProtoNet and FEAT methods. Fig. 8 shows the classification results of the proposed method and the comparative method. It can be seen that the OSNN method misclassified most of the unknown classes into the Suaeda Salsa class, and FEAT misclassified the unknown classes into the Tidal Flat Reed class. The ProtoNet method predominantly misclassified Tidal Flat Reed as Tamarix. However, the method proposed in this article can map features to a more discriminative space, resulting in classification results that closely align with the truth map, thus achieving the best classification effect.

Table VI presents the classification accuracy of the YRE_2 dataset. Similar to the UH datasets, our method is still optimal, which illustrates the effectiveness of the method. The proposed method performs domain adaptation through DJPA, which enhances the discriminability of features while aligning source and target domain features, thereby achieving better results. Because of the limited spectral information, OSC-SCL-M demonstrates comparatively lower classification accuracy. The few-shot closed-set classification methods ProtoNet and FEAT did not achieve ideal classification results; in particular, the recognition accuracy for unknown classes was significantly lower than the open-set few-shot classification methods SnaTCHer and Ours methods. Snatcher and Ours methods outperform OSC-SCL-M, indicating that cross-domain open-set few-shot classification methods can effectively learn discriminative features from HS data. Fig. 9 shows that the OSNN, FEAT, and ProtoNet methods have poor classification results for unknown classes, and most of the unknown classes are mistakenly classified into the Suaeda Salsa class. In the Snatcher method, the unknown class and the Tidal Flat Reed class were seriously misclassified, and the proposed method achieved the best classification effect. Therefore, the effectiveness of the method is further verified by comparing the classification result images of the proposed method and the contrasting method. Comparing Tables V and VI, it can be seen that the classification accuracy of the two Yellow River Estuary datasets is similar. In addition, by comparing Figs. 8 and 9, we can observe that the YRE_2 dataset produced more continuous patches than the YRE_1 dataset. This difference arises because the MS image in the YRE_1 dataset was simulated using GF5 data, with a spatial resolution consistent with that of GF5. In contrast, the MS images in the YRE_2 dataset were captured by WFV-GF1, which has a higher spatial resolution than the simulated MS images in the YRE_1 dataset, resulting in more continuous patches.

Table VII presents the classification results of the Harbin dataset. As indicated, our method achieved the best results, with the OA of 73.15%, which is significantly higher than that of other methods and approximately 3.21% higher than the OA of the baseline method, OSC-SCL-H. In addition, our method demonstrates significantly better classification results for unknown classes compared to other methods. Among the baseline methods, OSC-SCL-H outperforms OSC-SCL-M because the HS spectrum contains more spectral information, making it easier to distinguish different types of objects. Among the comparison methods, the OSNN method performs the worst, as it only considers open-set classification through distance ratios and does not account for the scenario of limited samples. The evaluation index OS^* in the ProtoNet and FEAT methods is significantly better than OS, indicating that these two methods perform better on known classes than on unknown classes. This is primarily because these methods only address the classification problem in a closed set with limited samples, without considering unknown samples. The overall performance of the SnaTCHer method is better than that of ProtoNet and FEAT, as the SnaTCHer method accounts for both FSL and unknown classes. To further verify the effectiveness of the proposed method, Fig. 10 illustrates the classification results of the Harbin dataset. It is evident from the figure that the OSNN method produces very noisy classification results, with the worst overall performance. The OSC-SCL-H method outperforms OSC-SCL-M. In the ProtoNet and FEAT methods, most unknown classes are mistakenly classified as known classes, such as Scrub or Trees. While SnaTCHer is better at classifying the unknown classes than ProtoNet and FEAT, it still misclassifies some Farmland as Scrub. The figure shows that the proposed method provides the best classification results for unknown classes, and the overall classification outcome is the closest to the ground truth. This demonstrates that the method proposed in this article enhances the discriminative

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ability of the learned features, thereby achieving the best performance.

D. Alignment Performance

In order to further verify the effectiveness of the proposed method, t-distributed stochastic neighbor embedding (t-SNE) visualizations are used to display the feature maps extracted by the network. As shown in Figs. 11–14, the results of mapping the test parts of the four datasets to the 2-D space in different methods are shown. The original distribution of the MS image is very messy, and the feature map of the HS image is better than that of the MS image. As shown in Fig. 11, the feature map of the UH dataset is presented. In the FEAT method, it is evident that the unknown class and the commercial class are aliased. As depicted in Fig. 8, the unknown class on the right side of the FEAT classification result is mistakenly categorized into the commercial class. Similarly, in the ProtoNet method, the unknown class, commercial class, and highway class are all severely aliased, resulting in a very poor classification effect on unknown classes. In contrast, the method proposed in this article has the best distinction between each category and the unknown class, with only a few instances of aliasing observed in some classes. Consequently, the proposed method demonstrates the most effective recognition of unknown classes. Fig. 12 illustrates the t-SNE maps of the YRE_1 dataset. In the original space and most methods, unknown classes and other known classes are blended. Notably, in the FEAT and ProtoNet methods, the classes Reed and Tamarix are seriously aliased. As shown in Fig. 9, it can be seen from the classification result figure that the two methods are not ideal for the Reed and Tamarix category classification. The method proposed in this article effectively diminishes intraclass distance and enhances interclass distance through the adaptive joint similarity metric and SSFIE module, resulting in improved results. Fig. 13 shows the t-SNE maps of the YRE_2 dataset. In the proposed method, the mapping features of the Spartina Alterniflora class and Water class are more aggregated, and the distance between known classes and unknown classes is larger. In the SnaTCHer method, the feature map of the unknown class and the Tidal Flat Reed class is mixed together, resulting in the misclassification of the unknown class into the Tidal Flat Reed class in the classification map. Similarly, in the FEAT method, the unknown class and most classes are mixed, and the features of the Tidal Flat Reed class and Water class are partially mapped together. As shown in Fig. 10, the water class in the FEAT classification map is mistakenly classified into the Tidal Flat Reed class. In the ProtoNet method, there is a significant overlap between the Suaeda Salsa and unknown classes with the Tidal Flat Reed class. From the classification map, it is evident that most of the classes in the lower area of the YRE_2 dataset are mistakenly classified into the Suaeda Salsa class. Fig. 14 displays the t-SNE maps of the Harbin dataset. In the original space, the features of MS images exhibit more severe aliasing compared to HS images, resulting in poorer classification performance for MS images. In the ProtoNet method, it can be seen that the unknown class, Towns, and Scrub are aliased. From the classification map, it can be seen that the town in the central area is misclassified as the unknown class. In the FEAT and SnaTCHer methods, aliasing occurs among the unknown class, Scrub, Farmland, and Towns. In Fig. 10, it can also be seen that part of the Farmland class instances is misclassified as Towns class in the two methods. Compared with other methods, the approach proposed in this article more effectively aggregates features of the same class and reduces aliasing between the unknown class and other classes, thus achieving the best performance. Consequently, the t-SNE maps of the four datasets demonstrate that the proposed method successfully separates unknown classes from known classes and enhances the classification ability of the model.

E. Ablation Study

The framework proposed in this article mainly consists of four main components, which are FSL, SSFIE module, DJPA module, and BNM module. As shown in Table VIII, ablation studies were conducted on the four datasets by removing different components of the proposed method, resulting in a total of five variants: Ours-V0, which includes only the basic feature extraction part and FSL; Ours-V1, where the FSL component is removed from the model; Ours-V2, indicating the removal of the BNM module; Ours-V3, which deletes the DJPA component from the model; and Ours-V4, where the SSFIE module is removed from the model. Each experiment was conducted independently ten times. As depicted in Table IX, the averages and deviations of the three evaluation indicators, OA, Kappa, and HOS, are presented. The proposed method is higher than other ablation methods in all evaluation indicators, verifying the effectiveness of each part. Comparing the Ours-V0 and Ours-V1 methods, it is apparent that the FSL module plays a crucial role in few-shot classification. Removing the DJPA component in Ours-V3 results in a decrease in classification accuracy compared to Ours. This indicates that reducing the feature distribution difference between the source and target domains and improving the discriminability of features can improve the classification accuracy. In addition, the BNM module also contributes to improving the discriminability of features. Comparisons between Ours-V2 and Ours show improvements in OAs for the Houston dataset, YRE_1 dataset, YRE_2 dataset, and Harbin dataset by 0.88%, 1.13%, 1.11%, and 2.12%, respectively. Although these improvements are not as substantial as those achieved by ablating other modules, the BMN module has a positive impact on all four datasets. It is evident that the BMN module contributes to enhancing classification performance to a certain extent. Furthermore, comparing Ours and Ours-V4, we find that the SSFIE module proposed in this article effectively improves the recognition accuracy of both known and unknown classes. In summary, the integration of each module effectively enhances the accuracy of cross-domain few-shot HS–MS image classification.

IV. Conclusion

This article proposes an open-set HS–MS collaborative classification method under the condition of limited labeled samples. It primarily utilizes partially overlapping HS images to enhance the classification accuracy of large-scale MS

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images, effectively distinguishing known classes while identifying unknown classes. First, the SSFIE module is designed to interactively enhance spectral–spatial features of HS–MS images to provide richer feature representation. Second, in the FSL module, an adaptive joint similarity measurement criterion is designed so that the source domain and target domain can generate more discriminative features. The model aligns source and target domain data through DJPA, and the BNM module is introduced to enhance model discriminability. During the testing phase, the classification of known classes and unknown classes is achieved through the designed open-set classification module. Experiments on four HS–MS datasets demonstrate the superiority of the proposed method. In the experiment, compared to the open-set classification method OSNN, the few-shot classification methods ProtoNet and FEAT, and the open-set few-shot classification methods OSC-SCL and SnaTCHer, the proposed method significantly improves the classification accuracy of unknown classes while maintaining satisfactory accuracy for known classes. It achieves the highest evaluation indicators, including OS, HOS, OA, and Kappa, across all four datasets, demonstrating its effectiveness. In addition, ablation experiments and analysis further validate the effectiveness of each component of the proposed model. In future work, several key challenges can be addressed to improve the method. First, we will explore more effective techniques for matching HS and MS images to better combine their strengths. Second, robust solutions will be developed to reduce the impact of temporal differences in data acquisition and varying imaging conditions on classification accuracy. In addition, the approach can be integrated with ground sampling to support field survey campaigns in key areas.

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